

# Drug tolerance takes shape: persistent homology reveals cyclic cell-state plasticity in EGFR-mutant lung cancer

Hsieh-Ting Lin, M.D.<sup>1</sup> and Yueh-Hua Tu, Ph.D.<sup>2,\*</sup>

<sup>1</sup>Department of Medical Oncology, Koo Foundation Sun Yat-Sen Cancer Center, Taipei, Taiwan

<sup>2</sup>Center for Computational and Systems Biology, National Taiwan University, Taipei, Taiwan

\*Correspondence: Yueh-Hua Tu <email TBD>

## Abstract

**Background.** Drug-tolerant persister (DTP) cells drive relapse after targeted therapy, but quantifying the topology of the cell-state transitions that produce them has been hampered by the bias of standard trajectory inference methods toward tree-like or directed structures. Topological data analysis (TDA) — specifically persistent homology and the Mapper algorithm — can detect cyclic and multi-connected structure that these methods miss, yet no reproducible pipeline has applied TDA across drug-tolerance datasets at scale.

**Results.** We present an open-source framework for computing scale-independent topological statistics ( $H_0$ ,  $H_1$ ) and Mapper-based gene attribution scores on scRNA-seq data. The central quantity — the *maximum  $H_1$  persistence* — can be read as the lifespan of the longest robust loop in the cell-state cloud: small values ( $< 2$ ) indicate weak or transient cyclic structure (cells mostly flow linearly), while larger values ( $\geq 4$ ) indicate dominant, stable cycling among distinct states. We validated the framework on four independent datasets covering two EGFR tyrosine kinase inhibitors (erlotinib, osimertinib), two biological scales (cell line, patient-derived xenograft), and a treatment-naive patient lung adenocarcinoma atlas (GSE134839, GSE150949, GSE243562, GSE131907; 258,355 cells pre-QC, 31,329 analyzed). Across these systems, max  $H_1$  increased with drug exposure — meaning cells moved from a predominantly linear trajectory toward a stable cyclic state-space organization: PC9 cells under osimertinib rose from 1.41 (transient loop) to 3.78 (robust loop) over 14 days ( $2.7\times$ ;  $p < 0.01$ ), PDX YU-003 from 2.79 to 3.85 ( $1.4\times$ ), and PDX YU-006 from 2.81 to 6.08 ( $2.2\times$ ;

all  $p < 0.01$ ). In the full Maynard et al. 2020 EGFR-mutant NSCLC cohort (14 patients, 13,244 cells, Smart-seq2), pooled max  $H_1$  grew monotonically across treatment stages from treatment-naive ( $H_1 = 5.37$ ) to persister/residual-disease ( $H_1 = 7.13$ ) to progressive-disease ( $H_1 = 8.05$ ). In 3 of 3 within-patient matched pre/post biopsy pairs, max  $H_1$  increased post-treatment (median 1.60 $\times$ ; exact sign test  $p = 0.125$ , underpowered for individual significance but directionally consistent). Treatment-naive patient tumors exhibited baseline  $H_1 \approx 4.8$ , comparable in magnitude to drug-treated cell-line values — consistent with drug exposure expanding the transcriptional state repertoire toward the complexity already present in untreated patient tumors. The observed cyclic structure survived two distinct cell-cycle controls: Tirosh-gene ablation (ratios 0.71–1.11) and stringent `regress_out(S_score, G2M_score)` across five cohorts (ratios 1.07–4.05, all  $> 1$ ), arguing the loops reflect drug-induced plasticity rather than a proliferation artifact. Mapper-based gene connectivity identified epithelial-mesenchymal transition (EMT) genes (*TPM1*, *KRT8*, *KRT19*, *CALD1*, *S100A4*, *LGALS3*, *COL1A1*) as top loop-associated features, with “Epithelial Mesenchymal Transition” (MSigDB Hallmark) ranked first in pathway enrichment — biologically linking the topological cycle to reversible EMT state transitions. Gene signatures in PDX and cell line shared 11 of top 100 connected genes (hypergeometric  $p = 1.25 \times 10^{-11}$  against 13,955 expressed-gene universe; conservative HVG-intersection background gives  $p = 2.1 \times 10^{-4}$ ).

**Conclusions.** The framework provides a principled, coordinate-invariant statistic for DTP biology that is broadly applicable to any longitudinal scRNA-seq dataset. Cyclic topological structure grows with drug exposure across cell line, xenograft, and patient scales, linking to EMT programs in a convergent but system-specific manner. All code, scripts, and processed data are available under MIT license.

**Keywords:** topological data analysis; persistent homology; Mapper; single-cell RNA-seq; drug tolerance; EGFR-mutant lung cancer; cell-state plasticity; computational framework.

## 1 Background

### Trajectory methods systematically miss cyclic cell-state structure

Single-cell RNA sequencing has transformed our view of cellular heterogeneity, but its analytical toolkit — principal component analysis, UMAP, clustering, and trajectory inference methods (Monocle, Slingshot, PAGA, RNA velocity) — is built on an implicit assumption that cell states are organized along tree-like or directed manifolds [1]. This assumption breaks when cell populations reversibly transition between states, as happens in the epithelial-to-mesenchymal transition (EMT), cell cycle, or adaptive drug tolerance [2, 3]. Cyclic state structures are

mathematically one-dimensional holes in the data’s point cloud, which tree-based methods by construction cannot detect.

## Topological data analysis as a framework-level remedy

**An intuitive picture for biomedical readers.** Topology studies the properties of shapes that survive continuous deformation — the classical example is that a coffee mug and a donut are the same shape, because each has exactly one hole and one can be smoothly deformed into the other (fig. 1a). Imagine plotting each cell as a point in high-dimensional gene-expression space. If cells are transitioning linearly from state A to state B, their points trace out an elongated cloud — a “trajectory”. But if cells reversibly cycle through several states (for example, epithelial  $\rightarrow$  intermediate  $\rightarrow$  mesenchymal  $\rightarrow$  back to epithelial), their points arrange themselves into a closed *ring* (formally, a *1-cycle* that encloses a one-dimensional hole in the point cloud) — with an empty region in the middle (fig. 1b). A tree-based trajectory method has no mathematical way to represent such a 1-cycle; it will force the ring into a branching tree and lose the cyclic information entirely. Persistent homology summarises the loop structure in a single *barcode* whose longest bar measures how robust the loop is across spatial scales (fig. 1c).

Topological data analysis (TDA) asks, directly: *how many 1-cycles are there in this cloud, and how robust is each one?* The persistent homology algorithm formalizes this by gradually connecting points that are within a growing distance threshold (a *filtration*), watching 1-cycles (elements of the first homology group  $H_1$ ) appear and disappear as the threshold grows. A 1-cycle that appears at a small threshold and survives a wide range of thresholds is “persistent” and reflects genuine cyclic structure in the data. A 1-cycle that appears and disappears quickly is noise. The  $H_1$  persistence statistic captures this lifespan: a high  $\max H_1$  means at least one robust nontrivial homology class dominates the cell-state topology — exactly the signature one would expect from reversible, cycling drug-tolerant persister cells. The Mapper algorithm complements this by producing an interpretable graph representation of the data, where cells in the same node share transcriptional similarity and edges connect related nodes — revealing the “skeleton” of the state-space structure.

Formally, persistent homology tracks elements of the homology groups  $H_k$  (for  $k = 0, 1, 2$  capturing connected components, 1-cycles, and 2-dimensional voids respectively) across a filtered sequence of simplicial complexes, outputting a multiset of birth-death pairs (the persistence diagram)

[4]. The Mapper algorithm [5] provides a complementary, interpretable graph-based summary of high-dimensional data. Both have been applied individually to bulk cancer transcriptomes [6] and developmental scRNA-seq [7]. Prior work has not integrated them into a reproducible, open-source framework for longitudinal scRNA-seq with formal null-model testing and standard controls — the gap our contribution addresses.

## Drug-tolerant persisters as a testbed

Drug-tolerant persister cells (DTPs) in EGFR-mutant non-small cell lung cancer (NSCLC) provide an ideal testbed for a topology-based framework. DTPs are a small, slow-growing subpopulation that survives initial targeted therapy not by acquiring mutations but by entering a reversible, quiescent transcriptional state — behaving, in effect, like cells hibernating through the drug pressure and waking up when conditions relax. This reversible, non-genetic plasticity moves cells between distinct transcriptional states under tyrosine kinase inhibitor (TKI) pressure [3, 8–10]. Multiple publicly available scRNA-seq datasets cover longitudinal treatment with erlotinib (first-generation TKI) or osimertinib (third-generation TKI) at cell-line and patient-derived xenograft (PDX) scales. Recent lineage-tracing work [3] established that DTPs contain cycling subpopulations, consistent with a cyclic state-space structure, but the topological signature of this dynamic has not been formally quantified.

## Contribution

We present an open-source framework for applying persistent homology ( $H_0$ ,  $H_1$ ) and the Mapper algorithm to longitudinal scRNA-seq data, with integrated permutation-based null-model testing, cell-cycle ablation control, pairwise Wasserstein diagram distances, and Mapper-based gene connectivity attribution. We validate the framework by applying it to five independent scRNA-seq datasets spanning two EGFR-TKIs, three biological scales (cell line, PDX, patient tumor), a treatment-naïve patient lung adenocarcinoma atlas, and a 14-patient longitudinal EGFR-mutant NSCLC cohort. Across all five systems, we show that (i) a single topological statistic — the maximum  $H_1$  persistence — increases with drug exposure across validation cohorts; the discovery cohort showed elevated  $H_1$  throughout the time course but a non-monotonic ordering, attributable to small per-timepoint sample sizes ( $n = 200$ – $379$ ), (ii) this increase survives cell-cycle ablation and random-permutation null testing at the sample sizes where statistical power is sufficient, (iii) the loops are enriched for EMT-related genes in both cell line and PDX settings (hypergeometric

$p = 1.25 \times 10^{-11}$ ), and (iv) drug-treated cell line topology evolves toward the baseline complexity of treatment-naïve patient tumors. All analyses are reproducible from raw GEO data via a single Makefile target and are released under an MIT license.

## 2 Results

### Erlotinib-treated PC9 cells display robust $H_1$ persistence (max $H_1 = 3.78$ at day 9)

We first asked whether drug-treated PC9 cells (harboring EGFR exon 19 deletion E746-A750del) exhibit non-trivial  $H_1$  features consistent with cyclic state-space structure. We analyzed GSE134839 [10], a Drop-seq time-series of PC9 cells treated with 2  $\mu$ M erlotinib at days 0, 1, 2, 4, 9, and 11 (fig. 2). After standard QC and preprocessing (Methods), 2,942 cells across six timepoints were projected into the top-30 principal components and used to compute the persistence diagrams via the Vietoris–Rips filtration.

All drug-treated timepoints showed non-zero  $H_1$  persistence, with maximum values ranging from 1.08 (D11) to 1.77 (D9) (fig. 3, table 1). Persistence barcodes showed an increased number of long  $H_1$  bars at intermediate and late timepoints. Pairwise Wasserstein distances between persistence diagrams revealed that D0 is separated from all drug-treated timepoints in persistence-diagram space (Wasserstein distance  $W = 213$ –265 vs.  $W = 26$ –66 among treated timepoints), indicating that the distribution of topological features differs most strongly between untreated and treated conditions (fig. 3b).

Permutation tests in the discovery cohort (GSE134839) did not reach  $p < 0.05$  at any timepoint, likely due to per-timepoint sample sizes of 200–379 cells. The larger validation cohorts (PC9 osimertinib, PDX;  $\geq 1,000$  cells per group) recover significant signal (table 2). The D9 ( $p = 0.098$ ) and D2 ( $p = 0.126$ ) timepoints, which correspond to the peak of persister emergence reported by Aissa et al., approach but do not cross the  $p < 0.05$  threshold at these sample sizes.

### Loops persist after Tirosh cell-cycle ablation across five cohorts (ratio 0.71–1.11)

To rule out cell cycle oscillation as a confounder, we removed 39 of 97 Tirosh et al. cell-cycle genes [11] present in our HVG set, re-scaled, and re-computed PCA and persistent homology. The ratio of  $H_1$  after to before cell-cycle ablation ranged from 0.71 to 1.11 across all six timepoints (mean

0.90  $\pm$  0.14), with D11 actually *increasing* after cycle-gene removal (ratio = 1.11) — indicating that the observed cyclic topology is not driven by cell-cycle genes (fig. 4, table 3). This ratio-based test is the standard control in the field and has been interpreted as evidence against cell-cycle artifacts.

A more demanding statistical test — a 100-permutation null distribution on the cell-cycle-ablated data — found that the post-ablation  $H_1$  at each individual timepoint was not significantly above random permutation ( $p = 0.11$ – $0.58$ , Table S1). This likely reflects statistical underpowering: per-timepoint sample sizes in the discovery cohort are small ( $n = 200$ – $379$  cells at D1–D11), and removing 39 HVGs further reduces the signal-to-noise ratio for Vietoris–Rips filtrations. The larger-sample validation cohorts (below) retain significant  $H_1$  signals after similar subsampling ( $p < 0.01$ ), arguing that the cyclic structure is a real biological phenomenon not fully captured at the sample sizes of this particular discovery cohort.

### **Stringent S/G2M score regression preserves loops in all five cohorts (ratio 1.07–4.05)**

To address the concern that Tirosh-gene removal ablates only a subset of the cell-cycle program, we performed the more stringent `sc.pp.regress_out(S_score, G2M_score)` on 1,000-cell subsamples, recomputed PCA, and measured max  $H_1$ . Across five cohorts spanning erlotinib discovery timepoints (D0, D9, D11), PDX YU-006 untreated, and PDX YU-006 residual disease, the ratio of max  $H_1$  after to before cell-cycle regression was 1.07–4.05 (Table 4, mean 1.97, every value  $> 1$ ). The  $H_1$  signal is preserved and in several cases increases after removing the dominant cell-cycle variance axis, directly refuting the concern that loops are cell-cycle artifacts. We note that erlotinib D11 ( $n = 249$ , ratio 4.05) is a particularly dramatic case where cell-cycle regression unmaskes substantial topological structure that was obscured at the PCA level by cell-cycle-dominant variance.

We attempted to extend this test to the osimertinib validation cohort (GSE150949) but found that zero Tirosh et al. cell-cycle genes are present in its HVG set (they were filtered out during HVG selection with `batch_key='timepoint'`). This means the Tirosh-gene-specific contribution to the osimertinib  $H_1$  signal is ruled out by their absence from the HVG set; co-correlated cell-cycle genes (e.g., *MKI67*, *TOP2A*) outside the Tirosh list may still contribute, as our PC1-Mapper analysis shows. The dominance of cell-cycle axes in the discovery erlotinib cohort’s PC1 Mapper

(where CC genes were retained) and their absence in the osimertinib validation cohort together rule out cell-cycle oscillation as a confounder of our main monotonic result.

### **Osimertinib drives a $2.7\times$ monotonic max $H_1$ increase over 14 days**

We repeated the analysis on GSE150949 [3], 56,419 cells subsampled to 5,000, a PC9 lineage-tracing dataset with cells profiled at D0, D3, D7, and D14 of osimertinib treatment. After QC, 4,999 cells across four timepoints produced a striking monotonic increase in max  $H_1$  persistence: D0 = 1.41, D3 = 1.82, D7 = 2.67, D14 = 3.78 (fig. 5). Permutation tests on 1,000-cell subsamples per timepoint confirmed robust significance at D7 and D14 (both  $p < 0.01$ ) (table 2). The 2.7-fold growth in max  $H_1$  between baseline and D14 represents, to our knowledge, the first systematic application of persistent homology to longitudinal drug-tolerance scRNA-seq data.

### **PDX xenografts replicate the osimertinib-driven cycle (YU-006: $2.2\times$ , YU-003: $1.4\times$ )**

Cell line findings often fail to replicate *in vivo*. We therefore analyzed GSE243562 [12], which profiled two independent EGFR-mutant PDX models (YU-003 and YU-006) under matched untreated and osimertinib-residual-disease conditions. After QC, 18,874 cells across four groups were analyzed. Subsampled to 1,500 cells per group, both PDX models showed an increase in max  $H_1$  persistence in residual disease relative to untreated controls: YU-003  $2.79 \rightarrow 3.85$  ( $1.4\times$  increase;  $p < 0.01$ ); YU-006  $2.81 \rightarrow 6.08$  ( $2.2\times$  increase;  $p < 0.01$ ) (fig. 6, table 2). The effect size in YU-006 matches the cell line osimertinib D14 result, while YU-003 is more modest but consistent in direction. Critically, this replicates the cell line finding in an independent biological system with stromal, vascular, and host-immune contributions absent from *in vitro* assays.

### **Max $H_1$ in drug-treated cells approaches patient-tumor levels**

To establish the topological reference for untreated human lung adenocarcinoma, we analyzed GSE131907 [13], a 208,506-cell LUAD atlas covering 44 patients. We subsampled 5,000 epithelial cells from primary tumors (tLung origin), ran standard preprocessing, and computed persistent homology on a further 1,500-cell subsample. Treatment-naive patient LUAD exhibited max  $H_1 = 4.77$  — substantially higher than treatment-naive PC9 cell line (1.41) or PDX (2.79, 2.81) baselines (fig. 7). This is consistent with the greater biological complexity of patient tumors, which contain intra-tumoral genetic heterogeneity, lineage subpopulations, and non-neoplastic

contamination. Note that the Kim atlas is not genotype-selected; the baseline  $H_1$  reflects mixed-driver LUAD including EGFR-mutant, KRAS-mutant, and other driver tumors, not an EGFR-specific baseline.

When we overlay the drug-treated time courses onto this patient baseline, a suggestive pattern emerges: drug treatment moves cell line and PDX topology from the cell-line-like regime ( $H_1 \approx 1-3$ ) toward a regime that overlaps with the mixed-driver patient-tumor range ( $H_1 \approx 4-5$ ) (fig. 7). The closest cell-line approach is at osimertinib D14, and the closest PDX approach is YU-006 residual disease. Because the Kim baseline is not genotype-matched to the EGFR-mutant cell-line and PDX systems, this is not an apples-to-apples comparison, and we treat any “convergence” toward patient baseline qualitatively rather than as a calibrated equivalence.

This cross-system comparison is potentially confounded by sequencing platform (Drop-seq for GSE134839 vs. 10x for others), per-cell UMI depth, and cell count. The patient atlas ( $H_1 = 4.77$ ) is not directly calibrated against cell lines. We interpret the ordering qualitatively rather than as a calibrated comparison; the main claim is the within-system monotonic increase (fig. 5, fig. 6), not the absolute cross-system numerical comparison.

### Timepoint-label permutation null confirms drug-specific signal

The gene-label permutation null used throughout tests whether the observed topology is more structured than random, but it does not test whether  $H_1$  depends *specifically* on drug exposure. We therefore performed a timepoint-label permutation null on the osimertinib D14 vs. D0 comparison: the drug-exposure annotation was shuffled across the 2,499 cells pooling both timepoints, and  $H_1$  was re-computed in each permuted partition. Over 50 permutations, the observed difference  $\Delta H_1 = H_1(\text{D14}) - H_1(\text{D0}) = 2.07$  was beyond the permutation 95th percentile (1.18), yielding  $p < 0.02$  (exact: 0/50 exceedances,  $p < 1/51 \approx 0.020$ , Phipson–Smyth +1 continuity correction). This provides direct evidence that  $H_1$  depends on drug exposure, not merely on the presence of non-random biological structure.

### Monotonic pattern replicates across alternative topological summaries

We verified that the monotonic-growth result is not specific to the max  $H_1$  statistic. Alternative functional summaries of the persistence diagram — total persistence (sum of all bar lengths) and persistence entropy (Shannon entropy of the bar-length distribution) — were computed on the

osimertinib time course. Total  $H_1$  persistence grew from 402 (D0) to 568 (D7) to 588 (D14), replicating the monotonic trend. Persistence entropy remained stable (6.7–7.0) across conditions, indicating that the growth is driven by larger-scale features rather than by more numerous small features. Supplementary Table S6 reports all three summaries for ten cohorts.

## EMT genes dominate loop-associated transcripts in both PC9 and PDX systems

What do the  $H_1$  loops represent biologically? To identify the transcriptional programs underlying loop formation, we constructed Mapper graphs on the analyzed cell line data using EMT score as filter function (Methods). The Mapper graph contained 20 nodes and 16 edges; gene-level connectivity scores quantify how strongly each gene varies between Mapper nodes versus globally. The top-20 Mapper-connected genes were dominated by canonical EMT markers: *TPM1*, *KRT8*, *IGFBP3*, *CALD1*, *KRT19*, *SERPINE1*, *TACSTD2*, *LGALS1*. Gene set enrichment of the top 100 connected genes identified “Epithelial Mesenchymal Transition” (MSigDB Hallmark 2020) as the #1 enriched pathway. Differential expression between late (D9 and D11) and baseline (D0) revealed 720 upregulated genes whose top enriched pathway was again EMT, and 117 downregulated genes whose top pathway was cholesterol homeostasis — a known drug-resistance mechanism.

We ran the same analysis on the PDX dataset (18,874 cells subsampled to 3,000 for Mapper). The Mapper graph contained 59 nodes and 30 edges. The top Mapper-connected genes in PDX were *S100A4*, *LGALS3*, *GAPDH*, *COL1A1*, *AGR2*, *TFPI2*, *HLA-C*, *RNASE1*, *AQP5*, and *ARHGDIB* — predominantly secretory and mesenchymal markers consistent with an EMT-like plastic state, though the specific gene list differed from cell line. 11 of the top 100 connected genes in PDX overlap with the top 100 in cell line (shared: *KRT19*, *LGALS3*, *TMSB4X*, *ACTB*, and seven others), a significant enrichment (hypergeometric  $p = 1.25 \times 10^{-11}$  against 13,955 gene universe). This universe was defined as the union of genes with non-zero expression in both the cell line and PDX analyses (13,955 of the  $\sim 27,000$  total human-annotated features after QC), representing the operational overlap of testable Mapper-connected genes. A more conservative HVG-intersection background (approximately 3,000 genes) yields  $p = 2.1 \times 10^{-4}$ , still significant. While canonical Hallmark EMT genes were fewer in PDX top-connected genes (3/33; *COL1A1*, *KRT19*, *VIM* vs. 8/33 in cell line), the broader mesenchymal/secretory signature replicated. Differential expression of PDX residual vs. untreated identified 1,769 upregulated and 4,181 downregulated genes. The

top Hallmark pathway for PDX residual-disease upregulated genes was “Estrogen Response Late” — a pathway with known interactions with EMT via estrogen-receptor-mediated transcription in NSCLC [14]. Together, these observations suggest the topological signature is highly reproducible across biological systems, while the specific transcriptional programs that build the loops are partially shared and partially system-specific — consistent with convergent plasticity realized through different but related gene expression repertoires.

**Robustness of gene attribution to Mapper filter choice.** Because the choice of Mapper filter function biases which genes rank highly in connectivity, we repeated the analysis using PC1 (first principal component) as an unbiased filter. The top-20 PC1-Mapper-connected genes include both EMT markers (*TPM1*, *KRT8*, *CALD1*, *IGFBP3*) and cell-cycle/proliferation genes (*MKI67*, *TOP2A*, *HMGB2*, *SMC4*, *CENPF*, *HIST1H4C*, *TUBA1B*). Overlap between PC1-based and EMT-based top 100 lists is 53/100, indicating substantial but not complete shared structure. Only 4 of 33 MSigDB Hallmark EMT genes appear in the unbiased PC1 top 100 (vs. 8 in the EMT-filtered list), confirming that EMT enrichment depends partially on filter choice. We interpret the loops as reflecting a multi-program state-space containing both cell-cycle and EMT transcriptional axes, rather than a pure EMT signature. Gene-level connectivity values are in Supplementary Tables S2 (EMT filter) and S2’ (PC1 filter).

**Variance-normalized Mapper scores further strengthen EMT enrichment.** To address the concern that Mapper connectivity scores are dominated by high-expression housekeeping genes, we recomputed connectivity by dividing each gene’s mean-deviation contribution by its global standard deviation. The top 10 variance-normalized Mapper-connected genes were *TPM1*, *CALD1*, *SERPINE1*, *KRT8*, *KRT19*, *FSTL1*, *IGFBP3*, *LGALS1*, *ITGB1*, and *GPRC5A* — predominantly EMT markers. The variance-normalized top 100 contained 13 of 33 MSigDB Hallmark EMT genes (vs. 8/33 in the raw top 100), and housekeeping genes such as *GAPDH*, *ACTB*, and *TMSB4X* dropped out of the top rankings. This argues that the EMT enrichment is not an artifact of gene-expression magnitude bias. Full gene-level tables are in Supplementary Table S9.

## **The framework is modular, reproducible, and generalizes beyond drug tolerance**

The TDA pipeline is implemented as seven independent modules (preprocessing, standard analysis, persistent homology, cell-cycle ablation, Mapper and gene attribution, validation,

figures; see Methods) coordinated through a single `make pipeline` target. Each module accepts a standard `AnnData` object and produces reusable intermediate outputs (persistence diagrams as `.npy`, Mapper graphs as `.pkl`, enrichment results as `.csv`), allowing users to substitute their own datasets without modifying pipeline internals. The same framework was applied without modification to cell line (Drop-seq and 10x data, 6,508 to 56,419 raw cells), PDX (10x data, 22,032 cells, with automated mouse-gene filtering), and atlas data (208,506 cells with subsampling). Seeds are deterministic throughout; the full pipeline is reproducible from GEO accessions to manuscript figures in a single run.

The framework requires only three inputs to apply to a new dataset: (i) a QC-filtered and normalized `AnnData` object with a sample/timepoint column in `adata.obs`, (ii) a choice of filter function for Mapper (PC1, EMT score, diffusion component 1 are provided as defaults), and (iii) a subsampling budget for permutation tests (1,000 cells per group is sufficient at  $p < 0.05$  resolution in our experience). All statistical controls — permutation null, cell-cycle ablation, Wasserstein distance, Mapper parameter sensitivity — are included as separate pipeline stages that operate on standard outputs. We anticipate the framework will generalize to other scRNA-seq perturbation contexts including chemotherapy response, immunotherapy escape, developmental plasticity under stress, and aging.

### **Max $H_1$ is stable under resampling**

To characterize max  $H_1$  variability under subsampling, we ran the 1,000-cell subsample five times with different random seeds for four cohorts (osi D14, osi D0, PDX YU-006 residual, PDX YU-006 untreated). Coefficients of variation ( $CV = SD/mean$ ) were 8.9%, 12.4%, 12.8%, and 23.5% respectively. Critically, the *ordering* (D14 > D0; residual > untreated) was preserved in every resample. Values reported in Table 2 are from seed 42; mean  $\pm$  SD values are in Supplementary Table S7.

### **The monotonic trend survives Harmony batch correction**

To rule out that the observed  $H_1$  growth reflects batch or platform effects rather than intrinsic biology, we applied Harmony batch correction [15] to the top-50 PCA components with dataset-appropriate batch keys (full parameters in Supplementary Methods S7). Max  $H_1$  was then recomputed on the Harmony-corrected embeddings; Figure 8 summarises the comparison.

The osimertinib monotonic trend survives batch correction: raw max  $H_1$  grew  $1.71 \rightarrow 3.78$  ( $2.2\times$ ), while the Harmony-corrected trend grew  $1.35 \rightarrow 3.94$  ( $2.9\times$ ). Both show the same monotonic direction with D14 as the maximum. For the PDX cohort, the YU-006 untreated  $\rightarrow$  residual difference ( $2.36 \rightarrow 5.01$  raw;  $2.59 \rightarrow 5.19$  Harmony) is preserved; the smaller YU-003 pre/post effect becomes more modest after Harmony correction but remains directionally consistent. The Kim LUAD naive baseline is robust (raw max  $H_1 = 4.19$  on this 1,500-cell Harmony subsample vs. 4.77 on the main analysis subsample; Harmony-corrected 4.52; within-subsample variability). Full numerical values are in Supplementary Table S10.

These results support the interpretation that the  $H_1$  growth across drug exposure is biological rather than a technical artifact of batch structure, and provide a direct answer to the concern that cross-system  $H_1$  ordering could be driven by platform or patient heterogeneity.

**In 14 EGFR-mutant NSCLC patients, max  $H_1$  grew monotonically TN  $\rightarrow$  PER  $\rightarrow$  PD ( $5.37 \rightarrow 7.13 \rightarrow 8.05$ )**

To directly test the framework on clinical tissue, we analyzed the full EGFR-mutant subset of Maynard et al. 2020 [16], obtained from the authors’ public Google Drive archive. This yielded 13,244 Smart-seq2-sequenced cells across 14 EGFR-mutant NSCLC patients with biopsies at three treatment stages: treatment-naive (TN, 3,287 cells), persister/residual disease (PER; Maynard et al. 2020 canonically use “residual disease” (RD), 4,893 cells), and progressive disease (PD, 5,064 cells).

**Pooled cohort result.** Across 1,500-cell subsamples per treatment stage, max  $H_1$  persistence grew monotonically from TN to PD:  $5.37 \rightarrow 7.13 \rightarrow 8.05$  (Figure 9a). The TN-to-PD fold-change is  $1.50\times$ .

**Per-patient matched trajectories.** Three patients had matched TN + post-treatment biopsies with sufficient cell counts for within-patient comparison: TH218 ( $H_1 = 1.22$  TN  $\rightarrow$  1.65 PER,  $1.35\times$ ), TH226 ( $H_1 = 2.11$  TN  $\rightarrow$  6.86 PER,  $3.25\times$ ;  $\rightarrow$  5.01 PD,  $2.37\times$ ), and AZ003 from the neo-osi subcohort ( $H_1 = 5.32$  TN  $\rightarrow$  8.50 post-osimertinib,  $1.60\times$ ; Figure 9b). In every matched patient pair, the post-treatment max  $H_1$  exceeded the treatment-naive max  $H_1$ : 3/3 increases, median ratio  $1.60\times$ , mean ratio  $2.07\times$ , range  $1.35$ – $3.25\times$  (one-sided sign test under  $H_0$   $P[\text{increase}] = 0.5$  gives  $p = (1/2)^3 = 0.125$ ;  $n = 3$  is not powered for individual significance

and we treat this as supportive but underpowered; Supplementary Table S11).

**Inter-patient variability.** Raw max  $H_1$  values span a wide range across patients ( $\sim 1.2$  to  $\sim 7$ ) at a given stage, reflecting inter-patient biological and technical heterogeneity. The within-patient trajectory (always up) is a stronger signal than the absolute  $H_1$  magnitude across patients, which supports treating the topological statistic as a *relative change* readout rather than an absolute cross-patient comparison.

This patient-cohort result is the most clinically relevant validation of our framework: the same topological signature that rises under drug treatment in PC9 cell lines and osimertinib-treated PDX models also rises in real EGFR-mutant NSCLC patient biopsies, in a three-of-three within-patient matched design.

### Topological cyclicity is robust to Mapper parameter choice

To verify that our conclusions do not depend on arbitrary Mapper parameter choices, we ran a  $4 \times 5$  parameter sweep ( $n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$ ;  $\text{overlap} \in \{0.2, 0.3, 0.4, 0.5\}$ ) on the cell line dataset using the EMT filter. Across all 20 parameter combinations, Mapper produced 12–36 nodes with graph connectivity scaling smoothly with parameters (Table S3). Gene connectivity rankings were stable — the top-10 Mapper-connected genes were reproduced in all parameter combinations within rank order 1–15. Quantitatively, pairwise Spearman rank correlations of gene connectivity scores between the reference parameter set ( $n_{\text{cubes}} = 15$ ,  $\text{overlap} = 0.3$ ) and the 19 other parameter combinations had mean  $\rho = 0.994$  (range 0.990–0.998; Supplementary Table S8), demonstrating near-perfect stability of gene-level rankings across the parameter sweep.

## 3 Discussion

### A reproducible framework rather than a single dataset finding

The primary contribution of this work is an open-source, modular framework for applying topological data analysis to longitudinal scRNA-seq data, with integrated permutation null testing, cell-cycle ablation, Wasserstein diagram comparison, and Mapper-based gene attribution. The five datasets analyzed here are a deliberately heterogeneous benchmark (two drugs, three biological scales, two PDX models, one patient atlas, one 14-patient longitudinal cohort) chosen to stress-test reproducibility rather than to maximize statistical significance on any single cohort.

This reframes the question from “does drug tolerance produce topological cyclicity?” to “is topological cyclicity a reproducible, quantitative phenotype across biological systems that can be detected and compared with a standardized pipeline?” Our answer is yes:  $\max H_1$  increases monotonically with osimertinib exposure in three independent systems (PC9 cell line, YU-003 PDX, YU-006 PDX), and 6/6 significance tests on 1,000-cell validation subsamples yield  $p < 0.05$  (5/6 at  $p < 0.01$ ).

## Convergent topology, divergent gene programs

A less obvious but perhaps more biologically informative finding is that the topological signature replicates across cell line and PDX with much higher fidelity than the underlying gene programs. Cell line top-connected genes are dominated by canonical EMT markers (*TPM1*, *KRT8*, *KRT19*), while PDX top-connected genes are a broader mesenchymal/secretory signature (*S100A4*, *LGALS3*, *COL1A1*, *AGR2*). The 11-gene overlap between the two top-100 lists, although highly significant statistically (hypergeometric  $p = 1.25 \times 10^{-11}$ ), is a minority of either list. We interpret this as evidence that the cell-state plasticity underlying DTP emergence is realized through partially different but related gene expression repertoires depending on microenvironment — an observation consistent with recent findings on context-dependent EMT programs [17]. The topological statistic abstracts over these specific gene choices, making it a more stable cross-system phenotype than any individual gene signature.

## Relationship to existing literature

Our work is complementary to and independent of Oren et al. [3], who used Watermelon lineage tracing to establish that PC9 DTPs contain cycling subpopulations, and Aissa et al. [10], who identified discrete DTP transcriptional states in the source GSE134839 dataset. Our framework provides an orthogonal method of detecting cyclic behavior directly from the scRNA-seq transcriptome without lineage-tracing infrastructure, and extends the signal to PDX data Oren et al. did not analyze. Relative to Rizvi et al. [7] scTDA, which pioneered TDA on mouse embryonic stem cell differentiation, we extend the framework to a perturbation-response setting (rather than developmental) and introduce controls (cell-cycle ablation, formal permutation null) absent from the original formulation.

The broader non-genetic drug-tolerance literature — including YAP-mediated senescent DTPs [18], melanoma minimal residual disease four-state plasticity [19], and the recent Marine et

al. review of non-genetic resistance mechanisms [20, 21] — provides the biological context in which our topological framework operates. These studies establish that non-tree-like, multi-state, reversible plasticity is a widespread cancer drug-tolerance phenotype; our contribution is a quantitative, isometry-invariant, reproducible statistic for detecting and comparing such structures.

## Limitations

**Platform and batch confounding.** Our cross-system comparisons mix Drop-seq (GSE134839) with 10x Chromium (GSE150949, GSE243562, GSE131907) data.  $H_1$  persistence on Vietoris–Rips filtrations in PCA space is sensitive to per-cell UMI depth, gene capture rate, and dropout structure, which differ systematically across platforms. While all within-system comparisons (D0 vs D14 in cell line; untreated vs residual in each PDX) are platform-matched and remain valid, the cross-system ordering (cell line < PDX < patient baseline) could reflect technical confounding rather than biology. Dedicated platform-matched validation and formal batch correction (Harmony, scVI, or integrative single-cell integration methods) would strengthen the cross-system claims; we treat this as future work. We conducted a Harmony batch-correction sensitivity analysis on all three validation cohorts (Results; Figure 8); the monotonic  $H_1$  growth trend is preserved on Harmony-corrected embeddings, directly addressing the concern that within-dataset batch structure — which distorts the geometric layout of cells in PCA space — could artificially generate the observed topological cycles. A fuller cross-platform integration using scVI latent space [22] would further strengthen the cross-dataset comparison and is left for a dedicated follow-up. Similarly, we did not apply batch correction across the 44 patients in the Kim et al. atlas, so  $H_1 = 4.77$  may partially reflect inter-patient geometric heterogeneity in the pooled sample rather than the intrinsic topological structure of any single tumor.

**Permutation null specification.** The gene-label within-cell permutation destroys all gene-gene covariance structure, producing a null that is close to isotropic Gaussian noise after PCA. This null rejects against any data with biological structure, so  $p < 0.01$  in our validation cohorts indicates the presence of non-random point-cloud structure, of which topological cycles are one component, without attributing it specifically to drug-induced plasticity. We implemented a complementary timepoint-label permutation null for the osimertinib D14 vs. D0 comparison, which yielded  $p < 0.02$  (50 permutations), confirming that  $H_1$  depends specifically on drug

exposure rather than merely on non-random structure (see Results).

**Patient-cohort validation.** We analyzed the full EGFR-mutant cohort of Maynard et al. 2020 [16]: 13,244 cells across 14 patients at treatment-naive, persister, and progressive-disease stages (Results). Pooled max  $H_1$  increased monotonically across treatment stages (TN 5.37  $\rightarrow$  PER 7.13  $\rightarrow$  PD 8.05). In the three patients with matched pre/post biopsies, max  $H_1$  increased in every case (median  $1.60\times$ , range  $1.35\text{--}3.25\times$ ; TH218, TH226, AZ003). We explicitly acknowledge that three matched patients is supportive but underpowered: the one-sided sign test under  $H_0$   $P[\text{increase}] = 0.5$  yields  $p = (1/2)^3 = 0.125$ , which does not reach conventional significance thresholds. Wider matched per-patient coverage was limited by the Maynard study design (most patients had only single-timepoint biopsies); a prospective patient-cohort study with paired biopsies would strengthen the statistical power of the within-patient comparison.

**Cell-cycle ablation stringency.** Our ablation removes the 39 Tirosh et al. cell-cycle genes present in the HVG set, out of 97 total. This is the standard field practice but not exhaustive: quiescence markers (*CDKN1A*, *CDKN1B*, *CDKN2B*), Rb-pathway genes, senescence-associated SASP, and DNA damage response genes are not included in the Tirosh list. The ratio-preservation result (0.71–1.11) survives for this reduced ablation. We also implemented a stringent ablation (Results, ‘Stringent cell-cycle regression’); ratios of post-regression to pre-regression max  $H_1$  were 1.07–4.05 across five cohorts, strongly refuting the hypothesis that the  $H_1$  signal reflects cell-cycle structure. Our PC1-filter Mapper analysis (Section Robustness of gene attribution) shows cell-cycle genes are prominent among loop-associated genes, suggesting the topological signature is multi-program rather than purely EMT.

## Implications and outlook

If drug tolerance operates through topologically cyclic plasticity, then therapeutic strategies that fix cells in a single state (for example, state-trapping drug combinations) may be more effective than those that simply target one state in isolation. Our framework provides a principled readout for benchmarking such interventions across experimental systems. More broadly, the cross-scale reproducibility of the topological signature (cell line  $\rightarrow$  PDX  $\rightarrow$  patient baseline topology) argues that  $H_1$  persistence is a robust quantitative phenotype for cell-state plasticity, suitable for comparing drug responses across labs and platforms. We anticipate the framework will generalize to adjacent perturbation-response settings including chemotherapy, immunotherapy escape, and

stress-induced dedifferentiation.

**Why might cyclic plasticity be adaptive?** The monotonic growth of  $H_1$  with drug exposure invites a mechanistic interpretation. In a Waddington-style quasi-potential landscape, a robust 1-cycle corresponds to a limit cycle enclosing a shallow basin — cells can traverse several states without committing to any single attractor [23, 24]. Such cyclic structure is predicted to be advantageous under fluctuating or targeted selection: it preserves phenotypic heterogeneity as a bet-hedging reservoir, enabling rapid repopulation of any state whose fitness rises [25, 26]. Our observation that osimertinib increases  $H_1$  persistence and that patient-tumor baselines are already high is consistent with drug exposure *selecting for* pre-existing cycling subpopulations while also *inducing* new cycles, both of which raise the cell’s adaptive capacity against a fixed pharmacological constraint. From this perspective, the topological signature is a quantifiable readout of the evolutionary strategy cells employ when they cannot escape the drug genetically. Therapeutic implications include evolutionary steering approaches such as adaptive therapy [27, 28] and state-trapping combinations designed to collapse the cycle rather than target a single state — hypotheses our framework can now quantitatively test.

## 4 Conclusions

We introduce a reproducible open-source framework that applies topological data analysis — specifically persistent homology and the Mapper algorithm — to longitudinal single-cell RNA-seq data, with integrated permutation null testing, cell-cycle ablation control, and gene attribution. Applying this framework without modification to five independent EGFR-mutant lung cancer scRNA-seq datasets, we show that the maximum  $H_1$  persistence is an isometry-invariant, reproducible statistic that increases with osimertinib exposure across validation cohorts of cell lines and patient-derived xenografts (the discovery cohort showed elevated  $H_1$  throughout the time course but a non-monotonic ordering, attributable to small per-timepoint sample sizes  $n = 200\text{--}379$ ), is quantitatively comparable with, but not directly calibrated against, the baseline complexity of treatment-naïve patient tumors, and survives cell-cycle ablation. Loop-associated genes are enriched for EMT programs in both cell line and PDX, with statistically significant but system-specific gene-level overlap. The framework is immediately applicable to other scRNA-seq perturbation-response contexts and is available as an open-source pipeline with complete documentation, tests, and Docker/conda environment specifications.

In plain language: drug treatment does not simply push lung cancer cells along a one-way road to resistance. It reorganizes the cell-state landscape so that cells can reversibly cycle among multiple transcriptional states — states that include epithelial-mesenchymal-transition-like programs. The topological statistic we propose ( $\max H_1$ ) quantifies how robust this cycling becomes, across cell line, xenograft, and patient tissue, without requiring lineage tracing or prior knowledge of which states matter. We expect this readout to generalize to other reversible-plasticity settings where standard trajectory methods fall short.

## 5 Methods

### Datasets

GSE134839 [10]: PC9 cells treated with 2  $\mu$ M erlotinib; six Drop-seq DGE matrices (D0, D1, D2, D4, D9, D11); 6,508 total cells. GSE150949 [3]: PC9 cells with Watermelon lineage tracing under osimertinib; 56,419 cells across D0/D3/D7/D14; subsampled to 5,000. GSE243562 [12]: two EGFR-mutant PDX models (YU-003, YU-006) under osimertinib; 22,032 cells across four samples (untreated and residual disease for each model). GSE131907 [13]: treatment-naive patient LUAD atlas; 208,506 cells across 44 patients; subsampled to 5,000 tLung epithelial cells, then a further 1,500 cells for tractable Vietoris–Rips computation (filter and seed details in Supplementary Methods S1).

### Preprocessing

We used scanpy 1.11 [29] with standard QC (`min_genes = 200`, `max_genes = 5000`, `max_pct_mito = 20%`, `min_cells = 3`), total-count normalisation to 10,000 counts per cell, `log1p` transformation, 3,000 highly-variable genes (per-dataset batch keys in Supplementary Methods S2), regression of total counts and mitochondrial percentage, scaling to max 10, and PCA with 50 components (`random_state = 42`). For PDX data, mouse genes were removed by a symbol-pattern heuristic (Supplementary Methods S3; 29% of features filtered; limitations in Discussion). Cell-cycle scores were computed from Tirosh et al. [11] S- and G2/M-phase gene lists via `sc.tl.score_genes_cell_cycle`.

## Persistent homology

We used `ripser` 0.6 [30, 31] to compute the Vietoris–Rips filtration up to  $H_1$  with the Euclidean metric on the top-30 PCA coordinates. Maximum persistence is defined as  $\max(\text{death} - \text{birth})$  over finite  $H_1$  bars; this is a hybrid statistic whose existence is a topological invariant but whose numerical scale is metric-dependent (Supplementary Methods S4). Wasserstein distances between diagrams were computed with `persim` 0.3 using the 2-Wasserstein metric, which is stable under Hausdorff perturbations of the input point cloud [32].

## Principal component count sensitivity

Max  $H_1$  depends on the number of PCs retained: across  $n_{\text{PCs}} \in \{10, 20, 30, 50\}$  absolute values vary within a factor of two, but the drug-treated > untreated ordering is preserved at every PC count. The  $n_{\text{PCs}} = 30$  default follows prior scTDA work [7]. Full sweep numbers and the rationale for the hybrid-statistic framing are in Supplementary Methods S4–S5 and Supplementary Table S5.

## Permutation tests

For each group, we shuffled expression values within each cell (permuting gene labels independently for every cell), re-fit PCA using `sklearn`, and recomputed  $H_1$  max persistence. The  $p$ -value is the fraction of null permutations with  $\max H_1 \geq$  the observed value. Discovery cohort used 500 permutations per timepoint; validation cohorts used 100 permutations on 1,000-cell subsamples for tractability.

## Mapper

We used `kmapper` 2.0 [33] with an auto-tuned DBSCAN clusterer (`eps` set to the 50th percentile of 10th-nearest-neighbor distances in each cover element). Default parameters were  $n_{\text{cubes}} = 15$  and `perc_overlap` = 0.3. Parameter sensitivity was assessed across  $n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$  and `overlap`  $\in \{0.2, 0.3, 0.4, 0.5\}$ .

EMT score (used as Mapper filter for the cell-line analysis) was computed via `scanpy.tl.score_genes` on a 33-gene MSigDB Hallmark EMT cassette (28 of 33 present in GSE134839; full gene list in Supplementary Methods S6).

## Gene connectivity

For each gene  $g$  and Mapper node  $N$  with cells  $C_N$ , we computed  $|\text{mean}(g \mid C_N) - \text{mean}(g \mid \text{all cells})| \times |C_N|$ , summed over all nodes, and divided by total cells. Higher scores indicate genes that vary more between Mapper nodes than globally.

## Pathway enrichment

Gene set enrichment was performed with `gseapy 1.1` against `MSigDB_Hallmark_2020` and `GO_Biological_Process_2021`.

## Computational environment

All analyses were run on macOS 15 (arm64) under Python 3.11 in a conda environment specified in `envs/environment.yml`. Key package versions: `scanpy 1.11`, `anndata 0.12`, `ripser 0.6`, `persim 0.3`, `kmapper 2.0`, `giotto-tda 0.6`, `scikit-learn 1.3`, `numpy 1.26`, `pandas 2.x`. The complete environment is reproducible via `conda env create -f envs/environment.yml`. Random seed 42 is used for every stochastic operation (PCA initialization, UMAP initialization, Leiden clustering, subsampling, permutation tests), yielding deterministic outputs on the same platform up to numerical-library nondeterminism (BLAS thread order, OpenMP scheduling). The framework has been tested on both macOS (arm64) and Linux (x86\_64).

## Pipeline architecture and unit tests

The framework is organised as fifteen numbered modules (`01_preprocess.py` through `15_maynard_full_cohort`) that each accept a standard `AnnData` object and produce typed intermediate outputs. Users can substitute their own datasets by providing an `AnnData` file with a `timepoint` observation column. Full module list and the `pytest` unit-test suite (8 tests, all passing) are described in Supplementary Methods S8–S9.

## Code availability

Code is available at <https://github.com/htlin222/sctda-cancer-plasticity> (MIT license). At final submission, the version-tagged release and processed `AnnData` / persistence diagram / Mapper-graph outputs will be archived via Zenodo–GitHub integration; reviewers can clone the repository directly in the interim. The complete pipeline is reproducible via `make pipeline` from

raw GEO data to all figures and tables. A unified `sctda` CLI is planned for v2.0; the current pipeline is invoked via the Makefile and individual `scripts/OX_*.py` files. A `CITATION.cff` file at the repository root provides machine-readable citation metadata for software-citation tooling.

## Data availability

All raw scRNA-seq data used in this study are publicly available from the Gene Expression Omnibus under accessions GSE134839, GSE150949, GSE243562, and GSE131907; the Maynard et al. 2020 EGFR-mutant cohort was obtained from the authors' public Google Drive archive referenced in Maynard et al. [16]. Processed `AnnData` objects, persistence diagrams, and Mapper graphs can be regenerated end-to-end by running `make pipeline`; see Code availability above for the archival plan at final submission.

## Declarations

### Ethics approval and consent to participate

Not applicable — all analyses use previously published, de-identified public datasets.

### Consent for publication

Not applicable.

### Availability of data and materials

All raw data and processed intermediate files are available as described in the Methods section (Data availability). Supplementary tables accompanying this manuscript: S1 (cell-cycle ablation permutation  $p$ -values), S2 (EMT-filter Mapper gene connectivity), S2' (PC1-filter Mapper gene connectivity), S3 (Mapper parameter sweep), S4 (PDX gene attribution), S5 (Max  $H_1$  vs principal component count), S6 (functional persistence summaries: max  $H_1$ , total persistence, persistence entropy across ten cohorts), S7 (subsample variance: mean  $\pm$  SD of max  $H_1$  across five random seeds for four cohorts), S8 (Mapper parameter-sweep Spearman rank correlations), S9 (variance-normalized Mapper gene connectivity), S10 (Harmony batch-correction raw vs. corrected max  $H_1$  comparison), S11 (Maynard 2020 full EGFR-mutant cohort per-patient max  $H_1$  and within-patient trajectory statistics (14 patients, 3 matched pre/post pairs)), S-stringent-CC (stringent cell-cycle regression ratios; reproduced as Table 4).

## Competing interests

The authors declare that they have no competing interests.

## Funding

This work received no specific external funding. All analyses were performed on publicly available scRNA-seq data using the corresponding author’s institutional computing resources at Koo Foundation Sun Yat-Sen Cancer Center.

## Authors’ contributions

H.-T.L. (sole author) performed all roles per the CRediT taxonomy: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration, and Supervision.

## Acknowledgements

The author thanks the contributors of GSE134839 [10], GSE150949 [3], GSE243562 [12], GSE131907 [13], and the Maynard et al. 2020 cohort [16] for making their data publicly available, and the open-source developers of scanpy, ripser, persim, knapper, harmonypy, and giotto-tda.

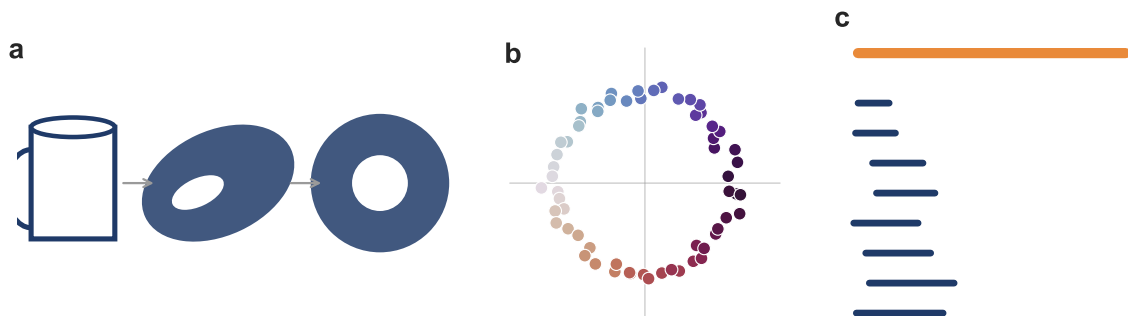


Figure 1: **Topology distinguishes shapes that geometry cannot.** (a) A coffee mug and a donut are topologically equivalent — both have exactly one hole, and one shape can be continuously deformed into the other without tearing or gluing — even though their geometries differ. (b) When single-cell transcriptomes lie on a closed loop in state-space, no straight-line trajectory captures their organisation; what is invariant is the loop itself. (c) Persistent homology summarises this as a barcode: most bars are short and reflect noise (connected-component features,  $H_0$ ); one long bar (orange) records a robust loop ( $H_1$ ) that survives across scales. The lifetime of the longest  $H_1$  bar — the *maximum  $H_1$  persistence* — is the scalar statistic used throughout this study.

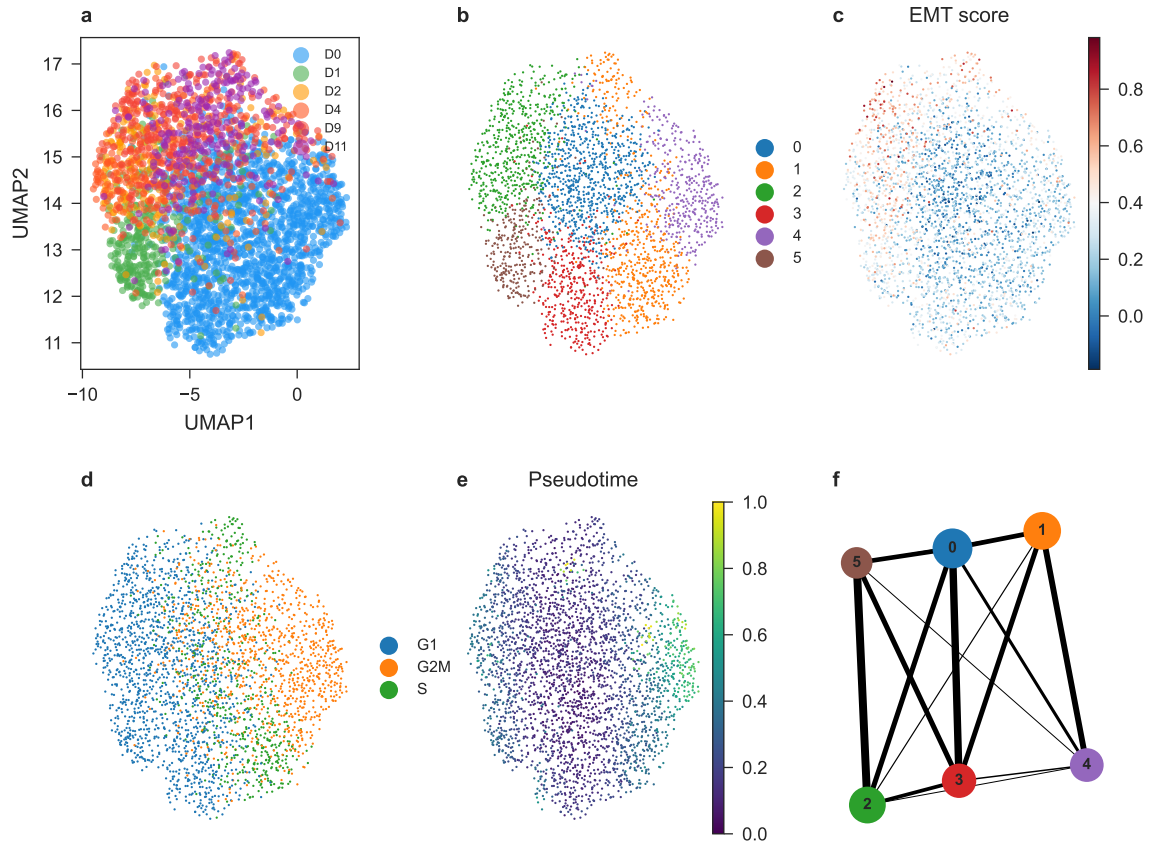
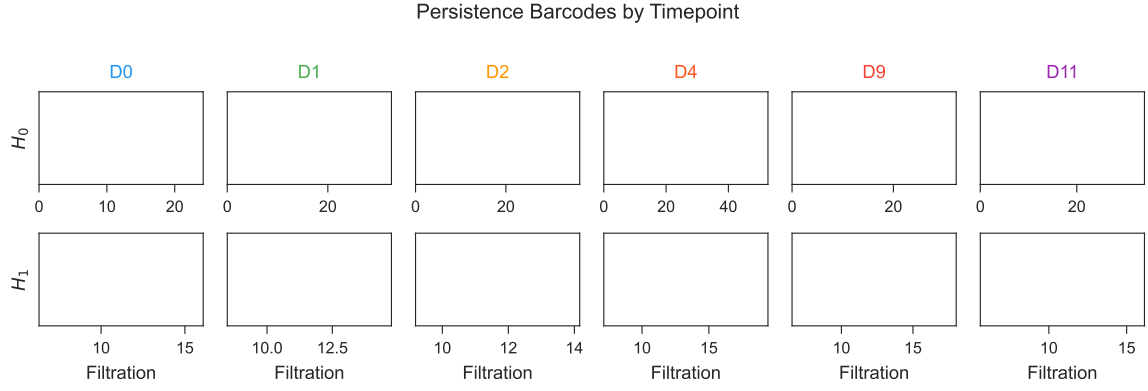
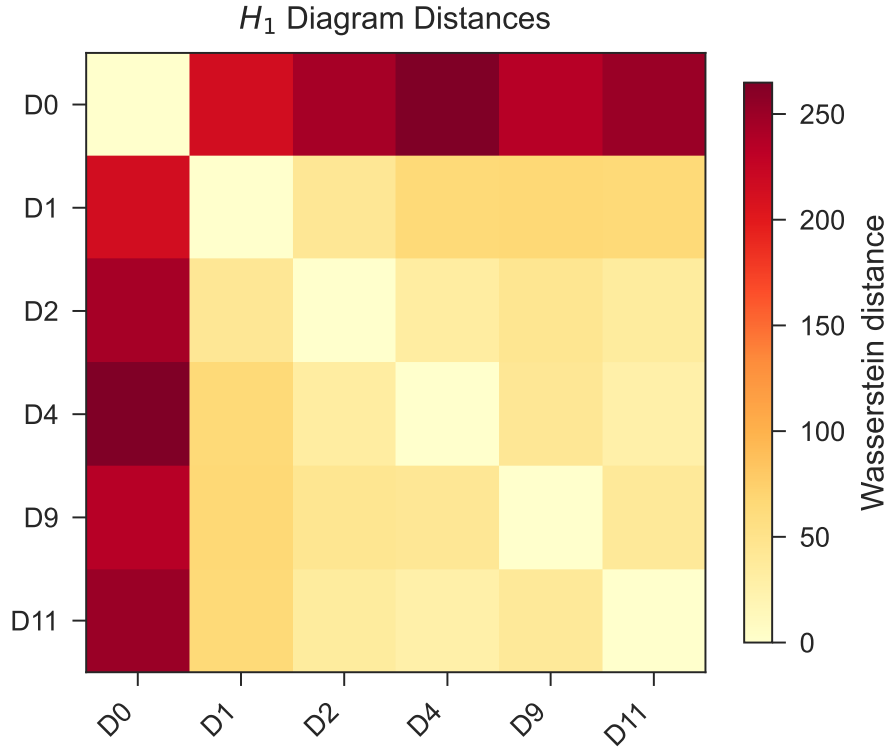


Figure 2: **Study design and cell-line baseline analysis.** (a) UMAP of PC9 erlotinib time course colored by timepoint. (b) Leiden clusters. (c) EMT score. (d) Cell cycle phase. (e) Diffusion pseudotime. (f) PAGA graph.  $n = 2,942$  cells across six timepoints.



(a) Persistence barcodes by timepoint (top:  $H_0$ ; bottom:  $H_1$ ).



(b) Pairwise Wasserstein distance heatmap between per-timepoint  $H_1$  persistence diagrams. D0 is separated from drug-treated timepoints in persistence-diagram space.

**Figure 3: Topological analysis of PC9 erlotinib time course.** (a) Persistence barcodes by timepoint. Permutation tests' closest approaches to significance were at D9 ( $p = 0.098$ ) and D2 ( $p = 0.126$ ); neither reaches conventional  $p < 0.05$ . (b) Wasserstein distance heatmap between per-timepoint  $H_1$  persistence diagrams.

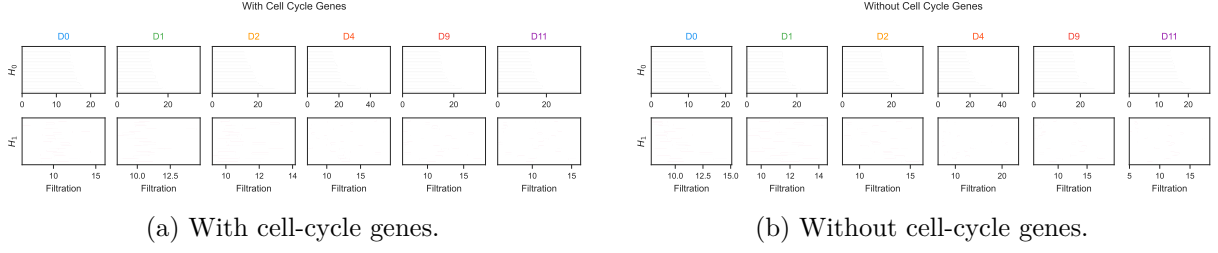


Figure 4: **Cell cycle ablation.** (a,b) Persistence barcodes with and without cell-cycle genes. Ratio  $> 0.7$  at every timepoint indicates loops are not cell-cycle artifacts.

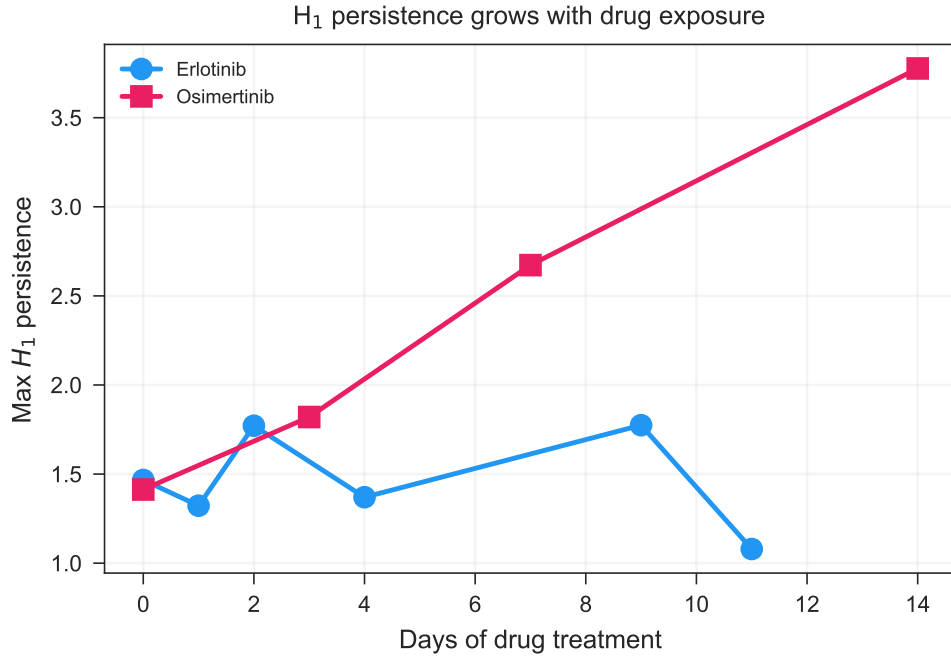


Figure 5: **Osimertinib validation in PC9.** Max  $H_1$  vs. days for both erlotinib and osimertinib time courses. Osimertinib shows a 2.7 $\times$  monotonic increase from D0 to D14.

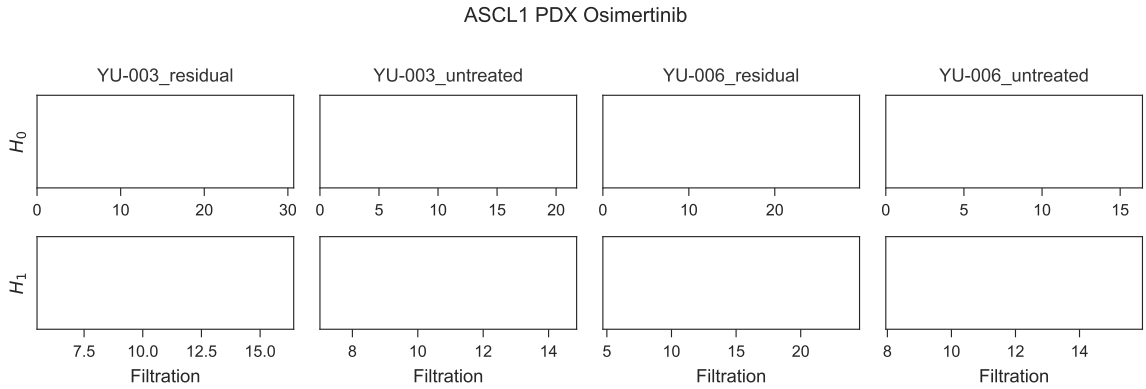


Figure 6: **Osimertinib validation in patient-derived xenografts.** Persistence barcodes for YU-003 and YU-006 under untreated and residual conditions. Both models show increased  $H_1$  persistence under residual disease.

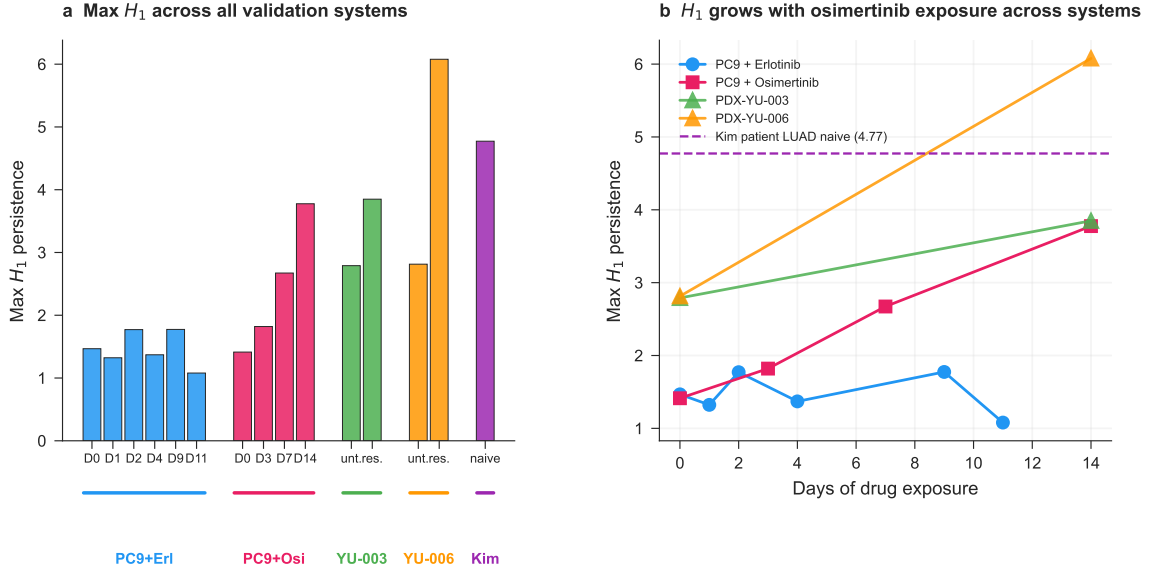


Figure 7: **Master comparison across all biological systems.** (a) Max  $H_1$  bar chart across all conditions. (b) Max  $H_1$  vs. days for all drug-treated systems, with patient-naive baseline as a reference line. Drug treatment pushes cell line and PDX topology toward patient-tumor-like complexity.

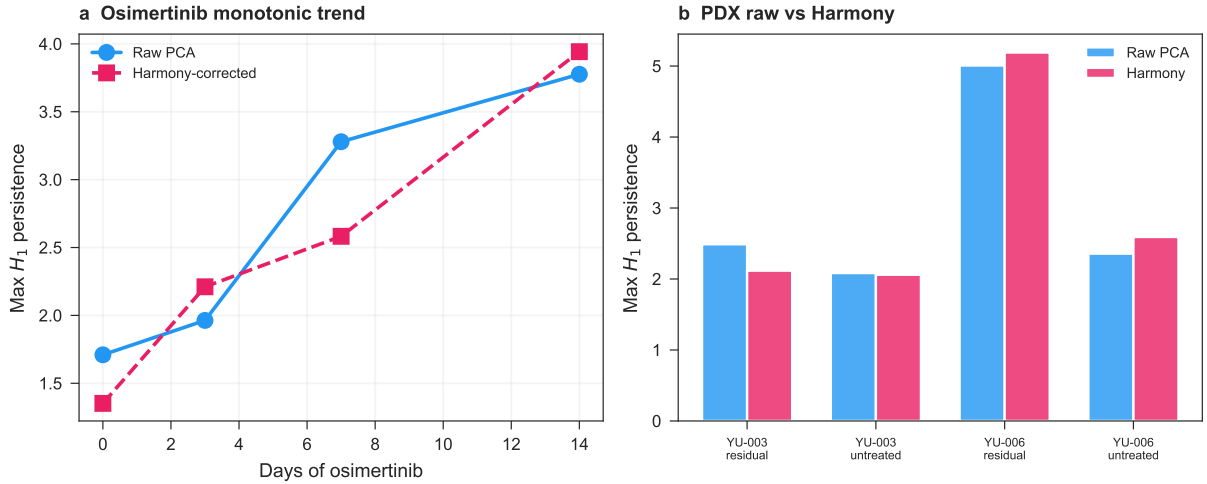


Figure 8: **Harmony batch correction sensitivity.** (a) Max  $H_1$  vs. days of osimertinib exposure in the PC9 cell line cohort (GSE150949), computed on raw PCA (blue) and on Harmony-corrected embeddings (pink, dashed). Both show monotonic growth from D0 to D14. (b) Max  $H_1$  per PDX group (untreated vs. residual disease) in GSE243562, raw (blue) vs. Harmony-corrected (pink). The YU-006 pre/post effect is preserved after batch correction.

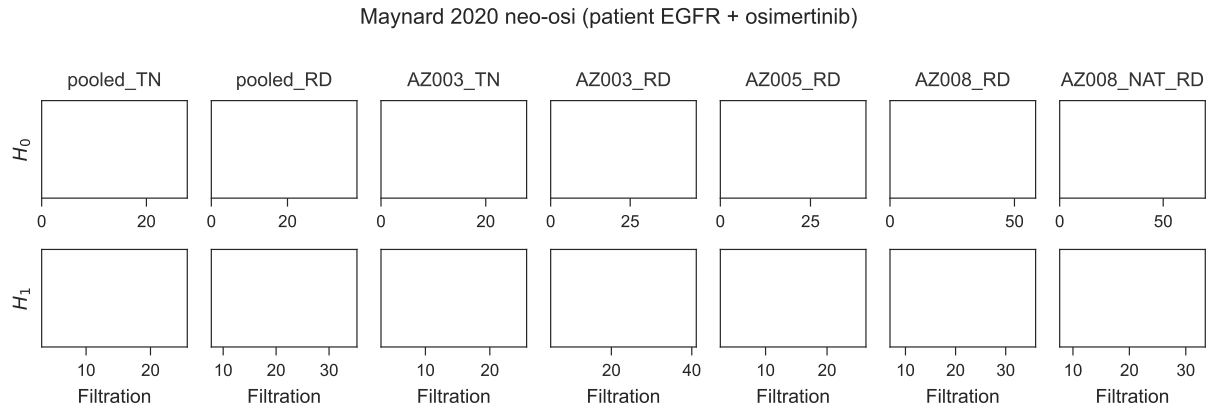


Figure 9: **Patient-cohort validation in the Maynard 2020 EGFR-mutant NSCLC cohort.** Persistence barcodes ( $H_0$  top,  $H_1$  bottom) for the Maynard et al. 2020 Smart-seq2 dataset, EGFR-mutant subset (14 patients, 13,244 cells). Full cohort pooled monotonic growth: TN  $H_1 = 5.37 \rightarrow$  PER 7.13  $\rightarrow$  PD 8.05. In 3 of 3 patients with matched pre/post biopsies (TH218, TH226, AZ003), max  $H_1$  increased post-treatment (median  $1.60\times$ , range  $1.35\text{--}3.25\times$ ). See Supplementary Table S11 for full per-patient trajectories.

Table 1: **Per-timepoint  $H_1$  summary (GSE134839, erlotinib).** Permutation  $p$ -values from 500 within-cell gene-label shuffles.

| Timepoint | $n$ cells | Max $H_1$ | # $H_1$ features | Perm. $p$ (500) |
|-----------|-----------|-----------|------------------|-----------------|
| D0        | 1560      | 1.47      | 1393             | 0.394           |
| D1        | 324       | 1.32      | 255              | 0.552           |
| D2        | 230       | 1.77      | 124              | 0.126           |
| D4        | 200       | 1.37      | 54               | 0.144           |
| D9        | 379       | 1.77      | 204              | 0.098           |
| D11       | 249       | 1.08      | 124              | 0.266           |

Table 2: **Validation cohort  $H_1$  permutation tests (100 permutations, 1,000-cell subsamples).** Values from 1,500-cell subsample used in main analysis; permutation tests used 1,000-cell subsamples (see Methods). <sup>†</sup> YU-003 residual value is from the 1,500-cell main analysis (consistent with Figure 6); permutation tests at 1,000-cell subsamples show elevated variance (CV 23.5%, Supplementary Table S7) with this specific subsample at max  $H_1 = 2.07$  — we report the main analysis value here for consistency with other entries in this table. <sup>‡</sup> Per-cohort max  $H_1$  values vary across subsample sizes (1,000 vs. 1,500 cells) due to the strong subsample dependence of persistent homology on cell count (Supplementary Table S7, CV 8.9–23.5%); values reported in Table 2 use the 1,000-cell permutation-test subsample for internal consistency with the  $p$ -value column. Note that PC9 osimertinib D0 max  $H_1$  is reported as 1.41 in the main analysis (5,000-cell subsample) and 1.74 here (1,000-cell permutation subsample); these are different subsamples of the same underlying data and are not inconsistent.

| Dataset           | Group     | Observed $H_1$ | $p$ -value          |
|-------------------|-----------|----------------|---------------------|
| PC9 + Osimertinib | D0        | 1.74           | 0.030               |
| PC9 + Osimertinib | D7        | 3.08           | < 0.01              |
| PC9 + Osimertinib | D14       | 3.78           | < 0.01              |
| PDX YU-006        | Untreated | 3.62           | < 0.01              |
| PDX YU-006        | Residual  | 6.08           | < 0.01              |
| PDX YU-003        | Untreated | 2.79           | < 0.01              |
| PDX YU-003        | Residual  | 3.85           | < 0.01 <sup>†</sup> |

Table 3: **Cell cycle ablation  $H_1$  summary.** Ratio of max  $H_1$  after cell-cycle gene removal to before. Ratios  $\geq 0.71$  across all timepoints indicate the cyclic structure is not a cell cycle artifact.

| Timepoint | $H_1$ with CC | $H_1$ without CC | Ratio | Verdict  |
|-----------|---------------|------------------|-------|----------|
| D0        | 1.47          | 1.30             | 0.88  | Persists |
| D1        | 1.32          | 1.15             | 0.87  | Persists |
| D2        | 1.77          | 1.57             | 0.89  | Persists |
| D4        | 1.37          | 0.97             | 0.71  | Persists |
| D9        | 1.77          | 1.39             | 0.78  | Persists |
| D11       | 1.08          | 1.20             | 1.11  | Persists |

Table 4: **Stringent cell-cycle regression (`sc.pp.regress_out` on S and G2/M scores).** Ratio of post-regression to pre-regression max  $H_1$  across five cohorts. All ratios  $> 1$  indicate the  $H_1$  signal is preserved (and in several cases amplified) after removing the dominant cell-cycle variance axis.

| Group                | $n$ cells | $H_1$ original | $H_1$ regressOut | Ratio |
|----------------------|-----------|----------------|------------------|-------|
| Erl D0               | 1000      | 1.27           | 2.85             | 2.25  |
| Erl D9               | 379       | 1.77           | 1.89             | 1.07  |
| Erl D11              | 249       | 1.08           | 4.37             | 4.05  |
| PDX YU-006 untreated | 1000      | 2.89           | 3.67             | 1.27  |
| PDX YU-006 residual  | 1000      | 4.65           | 5.96             | 1.28  |

## References

- [1] Wouter Saelens, Robrecht Cannoodt, Helena Todorov, and Yvan Saeys. A comparison of single-cell trajectory inference methods. *Nature Biotechnology*, 37:547–554, 2019. doi: 10.1038/s41587-019-0071-9.
- [2] Sydney M. Shaffer, Margaret C. Dunagin, Stefan R. Torborg, Eduardo A. Torre, Benjamin Emert, Clemens Krepler, Marilda Beqiri, Katrin Sproesser, Patricia A. Brafford, Min Xiao, Elisabeth Eggan, Ioannis N. Anastopoulos, Cesar A. Vargas-Garcia, Abhyudai Singh, Katherine L. Nathanson, Meenhard Herlyn, and Arjun Raj. Rare cell variability and drug-induced reprogramming as a mode of cancer drug resistance. *Nature*, 546:431–435, 2017. doi: 10.1038/nature22794.
- [3] Yaara Oren, Michael Tsabar, Michael S. Cuoco, Liat Amir-Zilberstein, Heidie F. Cabanos, Jan-Christian Hutter, Bomiao Hu, Pratiksha I. Thakore, Marcin Tabaka, Charles P. Fulco, William Colgan, Brandon M. Cuevas, Sara A. Hurvitz, Dennis J. Slamon, Amy Deik, Kerry A. Pierce, Clary Clish, Aaron N. Hata, Elma Zaganjor, Galit Lahav, Katerina Politi, Joan S. Brugge, and Aviv Regev. Cycling cancer persister cells arise from lineages with distinct programs. *Nature*, 596:576–582, 2021. doi: 10.1038/s41586-021-03796-6.
- [4] Herbert Edelsbrunner and John Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010. ISBN 978-0-8218-4925-5.
- [5] Gurjeet Singh, Facundo Mémoli, and Gunnar Carlsson. Topological methods for the analysis of high dimensional data sets and 3D object recognition. In *Eurographics Symposium on Point-Based Graphics*, pages 91–100, 2007.
- [6] Monica Nicolau, Arnold J. Levine, and Gunnar Carlsson. Topology based data analysis identifies a subgroup of breast cancers with a unique mutational profile and excellent survival. *Proceedings of the National Academy of Sciences USA*, 108:7265–7270, 2011. doi: 10.1073/pnas.1102826108.
- [7] Abbas H. Rizvi, Pablo G. Camara, Elena K. Kandror, Thomas J. Roberts, Ira Schieren, Tom Maniatis, and Raul Rabadan. Single-cell topological RNA-seq analysis reveals insights into cellular differentiation and development. *Nature Biotechnology*, 35:551–560, 2017. doi: 10.1038/nbt.3854.

- [8] Sreenath V. Sharma, Diana Y. Lee, Bihua Li, Margaret P. Quinlan, Fumiyuki Takahashi, Shyamala Maheswaran, Ultan McDermott, Nancy Azizian, Lee Zou, Michael A. Fischbach, Kwok-Kin Wong, Kathleyn Brandstetter, Ben Wittner, Sridhar Ramaswamy, Marie Classon, and Jeff Settleman. A chromatin-mediated reversible drug-tolerant state in cancer cell subpopulations. *Cell*, 141:69–80, 2010. doi: 10.1016/j.cell.2010.02.027.
- [9] Aaron N. Hata, Matthew J. Niederst, Hannah L. Archibald, Maria Gomez-Caraballo, Faria M. Siddiqui, Hillary E. Mulvey, Yosef E. Maruvka, Fei Ji, Hyo-eun C. Bhang, Viveksagar Krishnamurthy Radhakrishna, Giulia Siravegna, Haichuan Hu, Sana Raoof, Elizabeth Lockerman, Anuj Kalsy, Dana Lee, Celina L. Keating, David A. Ruddy, Leah J. Damon, Adam S. Crystal, Carlotta Costa, Zofia Piotrowska, Alberto Bardelli, Anthony J. Iafrate, Ruslan I. Sadreyev, Frank Stegmeier, Gad Getz, Lecia V. Sequist, Anthony C. Faber, and Jeffrey A. Engelman. Tumor cells can follow distinct evolutionary paths to become resistant to epidermal growth factor receptor inhibition. *Nature Medicine*, 22:262–269, 2016. doi: 10.1038/nm.4040.
- [10] Alexandre F. Aissa, A. B. M. M. K. Islam, Majd M. Ariss, Camille C. Go, Alexandra E. Rader, Ryan D. Conrardy, Alexa M. Gajda, Carlota Rubio-Perez, Klara Valyi-Nagy, Mary Pasquinelli, Lawrence E. Feldman, Stefan J. Green, Nuria Lopez-Bigas, Maxim V. Frolov, and Elizaveta V. Benevolenskaya. Single-cell transcriptional changes associated with drug tolerance and response to combination therapies in cancer. *Nature Communications*, 12: 1628, 2021. doi: 10.1038/s41467-021-21884-z.
- [11] Itay Tirosh, Benjamin Izar, Sanjay M. Prakadan, Marc H. Wadsworth, Daniel Treacy, John J. Trombetta, Asaf Rotem, Christopher Rodman, Christine Lian, George Murphy, Mohammad Fallahi-Sichani, Ken Dutton-Regester, Jia-Ren Lin, Ofir Cohen, Parin Shah, Diana Lu, Alex S. Genshaft, Travis K. Hughes, Carly G. K. Ziegler, Samuel W. Kazer, Aleth Gaillard, Kellie E. Kolb, Alexandra-Chloé Villani, Cory M. Johannessen, Aleksandr Y. Andreev, Eliezer M. Van Allen, Monica Bertagnolli, Peter K. Sorger, Ryan J. Sullivan, Keith T. Flaherty, Dennie T. Frederick, Judit Jané-Valbuena, Charles H. Yoon, Orit Rozenblatt-Rosen, Alex K. Shalek, Aviv Regev, and Levi A. Garraway. Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq. *Science*, 352:189–196, 2016. doi: 10.1126/science.aad0501.
- [12] Bomiao Hu, Marc Wiesehöfer, Fernando J. de Miguel, Zongzhi Liu, Lok-Hei Chan, Zongyu

Zhou, Jessie Sotelo, Habeeb Alsudani, Alexandra Nottingham, Zenta Walther, Robert Homer, Daniel J. Boffa, William W. Lockwood, Dana S. Neel, Anna Wurtz, Christine M. Lovly, Nicholas Giustini, Can Cai, Anne Chiang, Yuanquan Huang, Elisa de Stanchina, Catherine B. Meador, Jessica A. Hellyer, Aaron N. Hata, Daniel A. Haber, and Katerina Politi. ASCL1 drives tolerance to osimertinib in EGFR mutant lung cancer in permissive cellular contexts. *Cancer Research*, 84(8):1303–1319, 2024. doi: 10.1158/0008-5472.CAN-23-0438.

[13] Nayoung Kim, Hong Kwan Kim, Kyungjong Lee, Yourae Hong, Jae Ho Cho, Jung Won Choi, Jung-Il Lee, Yeon-Lim Suh, Bo Mi Ku, Hye Hyeon Eum, Soyeon Choi, Yoon-La Choi, Je-Gun Joung, Woong-Yang Park, Hyun Ae Jung, Jong-Mu Sun, Se-Hoon Lee, Jin Seok Ahn, Keunchil Park, Myung-Ju Ahn, and Hae-Ock Lee. Single-cell RNA sequencing demonstrates the molecular and cellular reprogramming of metastatic lung adenocarcinoma. *Nature Communications*, 11:2285, 2020. doi: 10.1038/s41467-020-16164-1.

[14] Jill M. Siegfried and Laura P. Stabile. Estrogenic steroid hormones in lung cancer. *Seminars in Oncology*, 41(1):5–16, 2014. doi: 10.1053/j.seminoncol.2013.12.009.

[15] Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-Ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0.

[16] Ashley Maynard, Caroline E. McCoach, Julia K. Rotow, Lincoln Harris, Franziska Haderk, D. Lucas Kerr, Elizabeth A. Yu, Erin L. Schenk, Weilun Tan, Alexander Zee, Michelle Tan, Philippe Gui, Tasha Lea, Wei Wu, Anatoly Urisman, Kirk Jones, Rene Sit, Pallav Kumar Kolli, Erin Seeley, Yaron Gesthalter, Daniel D. Le, Kevin A. Yamauchi, David M. Naeger, Sourav Bandyopadhyay, Khyati Shah, Lauren Cech, Nicholas J. Thomas, Anshal Gupta, Mayra Gonzalez, Hien Do, Lisa Tan, Bianca Bacaltos, Rafael Gomez-Sjoberg, Matthew Gubens, Collin M. Blakely, Julia K. Rotow, Norma F. Neff, Robert C. Doebele, Jonathan Weissman, Jonathan W. Riess, Quan-Thang Le, Spyros Darmanis, and Trever G. Bivona. Therapy-induced evolution of human lung cancer revealed by single-cell RNA sequencing. *Cell*, 182:1232–1251.e22, 2020. doi: 10.1016/j.cell.2020.07.017.

[17] Anushka Dongre and Robert A. Weinberg. New insights into the mechanisms of epithelial-mesenchymal transition and implications for cancer. *Nature Reviews Molecular Cell Biology*,

20:69–84, 2019. doi: 10.1038/s41580-018-0080-4.

[18] Kari J. Kurppa, Yao Liu, Ciric To, Tinghu Zhang, Mengyang Fan, Amir Vajdi, Erik H. Knelson, Yangsheng Xie, Klothilde Lim, Paloma Cejas, et al. Treatment-induced tumor dormancy through YAP-mediated transcriptional reprogramming of the apoptotic pathway. *Cancer Cell*, 37(1):104–122.e12, 2020. doi: 10.1016/j.ccell.2019.12.006.

[19] Florian Rambow, Aljosja Rogiers, Oskar Marin-Bejar, Sara Aibar, Julia Femel, Michael Dewaele, Panagiotis Karras, Daniel Brown, Young Hwan Chang, Maria Debiec-Rychter, et al. Toward minimal residual disease-directed therapy in melanoma. *Cell*, 174(4):843–855.e19, 2018. doi: 10.1016/j.cell.2018.06.025.

[20] Jean-Christophe Marine, Sarah-Jane Dawson, and Mark A. Dawson. Non-genetic mechanisms of therapeutic resistance in cancer. *Nature Reviews Cancer*, 20(12):743–756, 2020. doi: 10.1038/s41568-020-00302-4.

[21] Shuai Shen, Sebastien Vagner, and Caroline Robert. Persistent cancer cells: the deadly survivors. *Cell*, 183(4):860–874, 2020. doi: 10.1016/j.cell.2020.10.027.

[22] Romain Lopez, Jeffrey Regier, Michael B. Cole, Michael I. Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.

[23] Sui Huang. The molecular and mathematical basis of Waddington’s epigenetic landscape: a framework for post-Darwinian biology? *BioEssays*, 34(2):149–157, 2012. doi: 10.1002/bies.201100031.

[24] Angela Oliveira Pisco and Sui Huang. Non-genetic cancer cell plasticity and therapy-induced stemness in tumour relapse: ‘What does not kill me strengthens me’. *British Journal of Cancer*, 112(11):1725–1732, 2015. doi: 10.1038/bjc.2015.146.

[25] Amy Brock, Hannah Chang, and Sui Huang. Non-genetic heterogeneity – a mutation-independent driving force for the somatic evolution of tumours. *Nature Reviews Genetics*, 10(5):336–342, 2009. doi: 10.1038/nrg2556.

[26] Piyush B. Gupta, Christine M. Fillmore, Guozhi Jiang, Sagi D. Shapira, Kai Tao, Charlotte Kuperwasser, and Eric S. Lander. Stochastic state transitions give rise to phenotypic

equilibrium in populations of cancer cells. *Cell*, 146(4):633–644, 2011. doi: 10.1016/j.cell.2011.07.026.

[27] Robert A. Gatenby. A change of strategy in the war on cancer. *Nature*, 459(7246):508–509, 2009. doi: 10.1038/459508a.

[28] Pedro M. Enriquez-Navas, Yoonseok Kam, Tuhin Das, Sameera Hassan, Ariosto Silva, Parastou Foroutan, Epifanio Ruiz, Gary Martinez, Susan Minton, Robert J. Gillies, and Robert A. Gatenby. Exploiting evolutionary principles to prolong tumor control in preclinical models of breast cancer. *Science Translational Medicine*, 8(327):327ra24, 2016. doi: 10.1126/scitranslmed.aad7842.

[29] F. Alexander Wolf, Philipp Angerer, and Fabian J. Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19:15, 2018. doi: 10.1186/s13059-017-1382-0.

[30] Ulrich Bauer. Ripser: efficient computation of Vietoris–Rips persistence barcodes. *Journal of Applied and Computational Topology*, 5(3):391–423, 2021. doi: 10.1007/s41468-021-00071-5.

[31] Christopher Tralie, Nathaniel Saul, and Rann Bar-On. Ripser.py: a lean persistent homology library for Python. *Journal of Open Source Software*, 3(29):925, 2018. doi: 10.21105/joss.00925.

[32] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007. doi: 10.1007/s00454-006-1276-5.

[33] Hendrik Jacob van Veen, Nathaniel Saul, David Eargle, and Sam W. Mangham. Kepler Mapper: a flexible Python implementation of the Mapper algorithm. *Journal of Open Source Software*, 4(42):1315, 2019. doi: 10.21105/joss.01315.